Daniel Fang

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Education

Queen's University — Kingston, ON
B.A.Sc. in Electrical Engineering — Cumulative GPA: 3.44/4.30 — *Aug. 2024 – May 2028*

Experience

Low-Voltage Sub-Team Member — Oct. 2025 – Present
Queen's Racing Formula SAE — *Kingston, ON*

- Designed low-voltage schematics and PCB layouts in Altium for the vehicle's power distribution system, including the pedal box board's 5V (4.5A) and 12V (1A) fused power rails and 3.3V MCU regulation
- Evaluated PCB waterproofing and conformal coating options for exposed vehicle electronics; findings were adopted by the sub-team ahead of the controls design review

Projects

RISC-V RV32I Processor | *SystemVerilog, cocotb, Python, Icarus Verilog* — Jun. 2026 – Present

- Designed and integrated a single-cycle RV32I core in SystemVerilog — ALU, 32 × 32 register file, instruction decoder, control unit, program counter, and instruction memory — executing R-type programs end to end
- Closed functional coverage across all 10 RV32I ALU operations and 6 corner-case bins (zero result, shift-by-0/31, mixed-sign compare, extremal operands) over 3,000 constrained-random vectors checked against a Python reference model derived from the ISA specification
- Mutation-tested every module pre-commit; caught a decoder bit-slice bug (`rd[10:7]` instead of the spec's `[11:7]`) that two directed tests missed because their register values aliased under the mutation, exposed only by an all-ones stimulus vector
- Caught a self-written testbench reporting a false 3/3 pass with zero simulation time elapsed (a missing `await`); rebuilt verification practice around confirming every test fails on a deliberate mutation before trusting it to pass
- Synthesized the register file for the target Xilinx Artix-7 (Basys 3) in Vivado: 109 LUTs (0.5% of device), with the tool mapping the 32 × 32 array to distributed RAM rather than discrete flip-flops — a synthesis-level detail invisible in simulation

Smart Walking Stick | *ELEC 290 Design Project* — Sept. – Dec. 2025

- Owned test and validation on a 5-person, 12-week build, characterizing three HC-SR04 ultrasonic channels to ±2–3 cm accuracy
- Verified NEO-6M GPS accuracy to ±3–5 m over a 30-minute, zero-reset stress test; flagged a missed 30–60 ms latency target against spec
- Benchmarked results against the design specification in the final report, documenting above 90% obstacle detection

MeshStat | *ESP32, ESP-NOW, I2S, FreeRTOS — KingHacks* — Jan. 2026

- Placed runner-up, Mayor's Innovation Track, among all student teams at KingHacks
- Designed a versioned point-to-point wireless protocol between two ESP32 nodes over ESP-NOW — packed structs, per-packet sequence numbers, and send/receive acknowledgment callbacks confirming delivery on every transaction
- Eliminated data races between a receive interrupt handler and the main loop by guarding shared state with critical sections, reasoning explicitly about atomicity across concurrent execution contexts

Technical Skills

Languages: SystemVerilog, Verilog, C/C++, Python (cocotb, Pandas, FastAPI), SQL
Verification: cocotb testbenches, reference modelling, constrained-random stimulus, coverage closure, mutation testing
Tools & Hardware: Icarus Verilog, GTKWave, Vivado, Quartus, Basys 3 FPGA, Altium, Arduino, ESP32, I2C/SPI, oscilloscope, LTSpice, Git, Linux

Leadership & Interests

Basketball Captain, J.L. Crowe Senior Team (13-player roster) and Co-Founder, Youth Recreational Basketball League (∼40 players, 4 teams) | ARCT Diploma in Piano Performance, Royal Conservatory of Music